

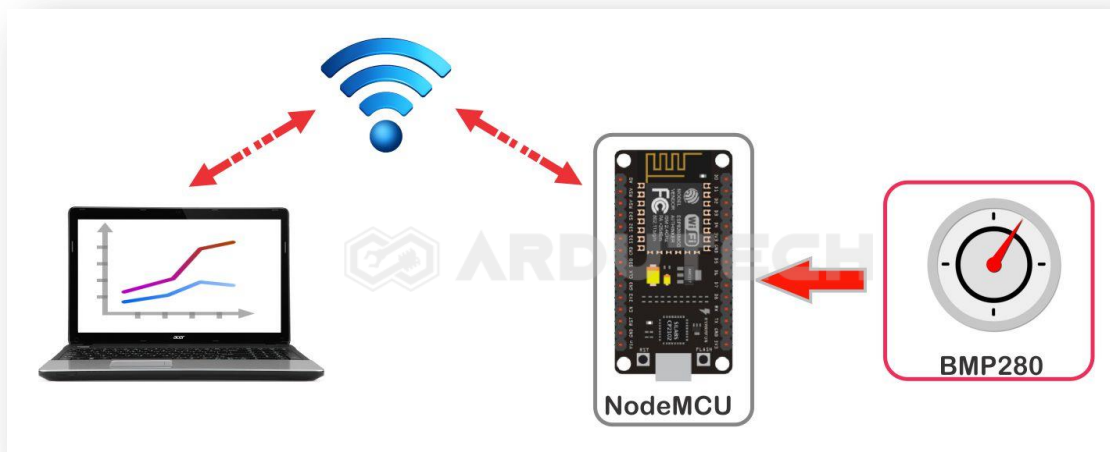
Monitoring Sensor BMP280 Web Page

Deskripsi

Monitoring sensor BMP280 (barometric & pressure sensor) dan hasilnya yang berupa temperature, pressure dan altitude dikirim melalui jaringan WiFi dan ditampilkan dalam Web Page.

Cara Kerja

BMP280 merupakan sensor berukuran fisik kecil dengan kemampuan mengukur tekanan barometrik (berat udara), temperature dan juga ketinggian (altitude). Sensor ini dapat berkomunikasi dengan model I2C maupun SPI & pada proyek ini nanti kita akan memakai komunikasi I2C yang memanfaatkan kaki SDA dan SCL.



Sensor BMP280 sudah dikemas dalam sebuah modul yang terdiri dari 6 pin seperti terlihat pada gambar berikut ini.



Spesifikasi :

- Altimeter : Mengukur ketinggian tempat dari permukaan air laut (0 – 9,2KM) dg akurasi - +1M
- Thermometer : Mengukur suhu udara/ruangan (Range : -40 sampai 85 C)
- Atmospheric Pressure Meter : Mengukur tekanan udara disuatu tempat (Range : 30,000Pa – 110,000Pa dg akurasi 12Pa)

- Komunikasi interface I2C dan SPI
- Address I2C : 0x76
- Akurasi relatif : 0.12hPa / 1 meter
- Akurasi absolut : 1hPa

Sensor BMP280 dihubungkan dengan NodeMCU V3 kemudian hasilnya dikirim melalui halaman web (web page) secara periodik. NodeMCU dan device yang memonitor (laptop/HP) harus satu jaringan WiFi (hotspot)

Kebutuhan Bahan

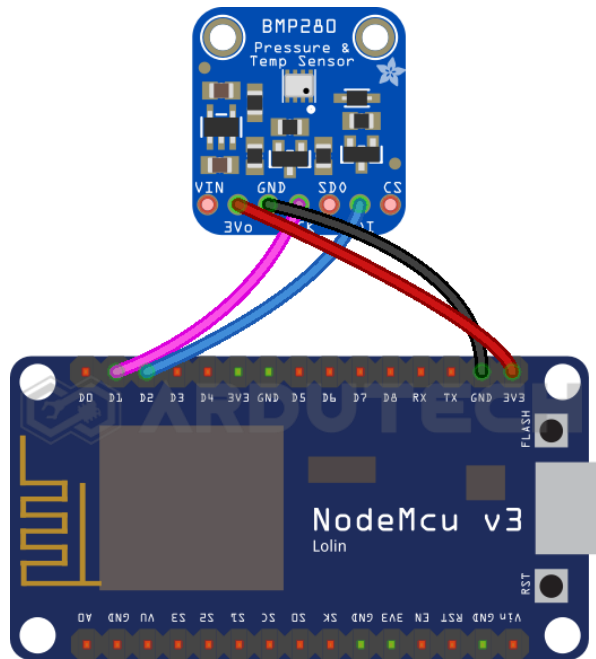
- NodeMCU V3
- Sensor Barometric Pressure BMP280
- Kabel konektor
- Kabel micro USB



Kebutuhan Software

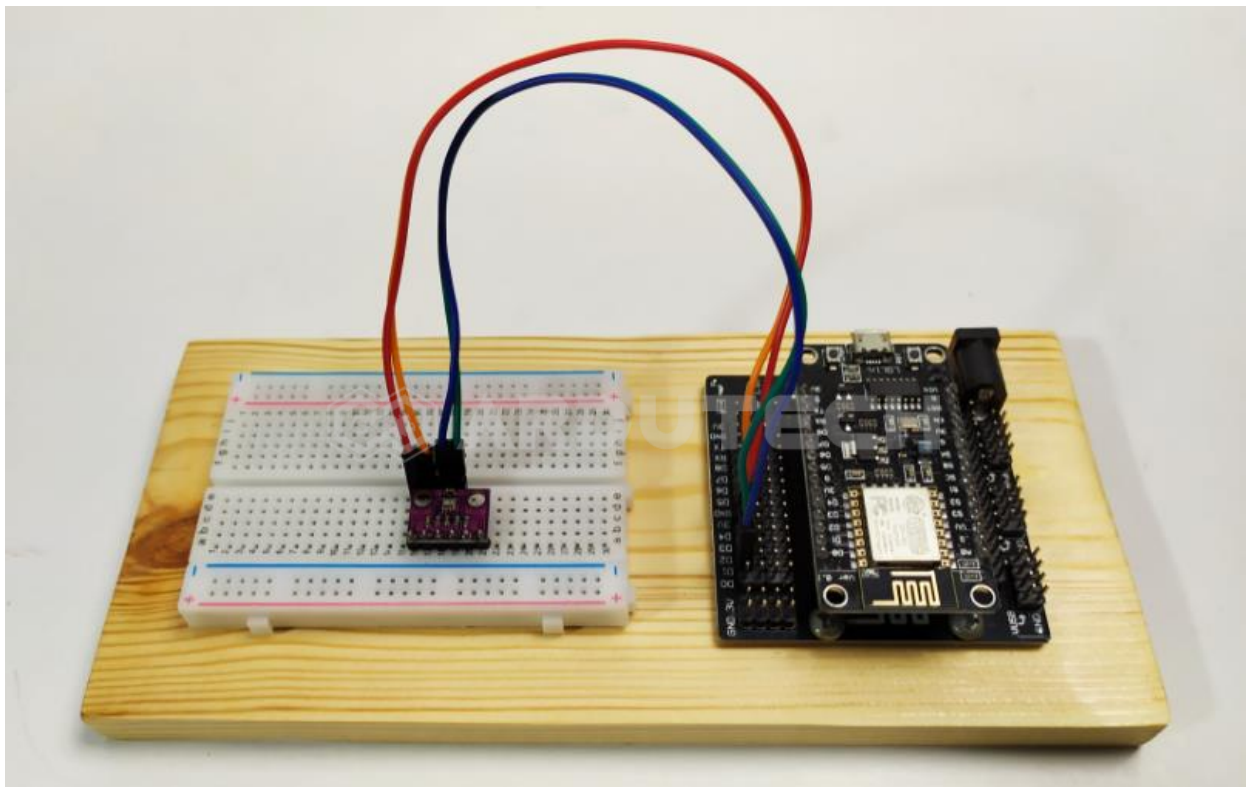
- Arduino IDE
- Source code html.

Rangkaian/Skematik



Koneksi NodeMCU dengan Sensor BMP280.

NodeMCU	BMP280
D1	SCL
D2	SDA
3V3	VCC
GND	GND



Program/Source Code

Program pada proyek ini memerlukan library :

- ESP8266WiFi.h
- Wire.h
- Adafruit_Sensor.h
- Adafruit_BMP280.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```

/*****
* Project Monitoring Sensor BMP280 via Webpage
* Board : NodeMCU ESP8266 V3
* Input : BMP280
* Output : Web page
* 99 Proyek IoT
* www.ardutech.com
*****/

#include <ESP8266WiFi.h>
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BMP280.h>
#define SEALEVELPRESSURE_HPA (1013.25)

Adafruit_BMP280 bmp;
float temperature, pressure, altitude;
//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
const char* ssid = "ArdutechWiFi"; // Nama Hotspot/WiFi
const char* password = "12345678"; // Password
WiFiServer server(80);
//=====

void setup() {
  Serial.begin(9600);
  delay(100);

```

```
bmp.begin(0x76);
Serial.println("");
Serial.print("Connecting to: ");
Serial.println(ssid);
//connect to your local wi-fi network
WiFi.begin(ssid, password);
//check wi-fi is connected to wi-fi network
while (WiFi.status() != WL_CONNECTED) {
  delay(1000);
  Serial.print(".");
}
Serial.println("");
Serial.println("WiFi connected..!");
Serial.print("IP address : "); Serial.println(WiFi.localIP());
server.begin();
Serial.println("HTTP server started");
delay(3000);
}

void loop() {
  temperature = bmp.readTemperature();
  pressure = bmp.readPressure() / 100.0F;
  altitude = bmp.readAltitude(SEALEVELPRESSURE_HPA);
  Serial.print("Temperature = ");
  Serial.print(temperature);
  Serial.println(" Celsius, ");
  Serial.print("Pressure = ");
  Serial.println(pressure);
  Serial.print("Altitude = ");
  Serial.println(altitude);
}
```

```
WiFiClient client = server.available();

client.println("HTTP/1.1 200 OK");

client.println("Content-Type: text/html");

client.println("Connection: close");

client.println("Refresh: 10");

client.println();

client.println("<!DOCTYPE HTML>");

client.println("<html>");

client.println("<head>");

client.println("<title>NodeMCU Weather Station</title>");

client.println("<meta name='viewport' content='width=device-width, initial-scale=1.0'>");

client.println("<link href='https://fonts.googleapis.com/css?family=Open+Sans:300,400,600' rel='stylesheet'>");

client.println("<style>");

client.println("html { font-family: 'Open Sans', sans-serif; display: block; margin: 0px auto; text-align: center;color: #444444;});");

client.println("body{margin: 0px;} ");

client.println("h1 {margin: 50px auto 30px;} ");

client.println(".side-by-side{display: table-cell;vertical-align: middle;position: relative;});");

client.println(".text{font-weight: 600;font-size: 19px;width: 200px;});");

client.println(".reading{font-weight: 300;font-size: 50px;padding-right: 25px;});");

client.println(".temperature .reading{color: #F29C1F;});");

client.println(".humidity .reading{color: #3B97D3;});");

client.println(".pressure .reading{color: #26B99A;});");

client.println(".altitude .reading{color: #955BA5;});");

client.println(".superscript{font-size: 17px;font-weight: 600;position: absolute;top: 10px;});");

client.println(".data{padding: 10px;});");

client.println(".container{display: table;margin: 0 auto;});");

client.println(".icon{width:65px}");

client.println("</style>");
```

```

client.println("</head>");
client.println("<body>");
client.println("<h1>NodeMCU Weather Station</h1>");
client.println("<div class='container'>");
client.println("<div class='data temperature'>");
client.println("<div class='side-by-side icon'>");
  client.println("<svg enable-background='new 0 0 19.438 54.003'height=54.003px id=Layer_1
version=1.1 viewBox='0 0 19.438 54.003'width=19.438px x=0px xml:space=preserve
xmlns=http://www.w3.org/2000/svg xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path
d='M11.976,8.82v-
2h4.084V6.063C16.06,2.715,13.345,0,9.996,0H9.313C5.965,0,3.252,2.715,3.252,6.063v30.982");
  client.println("C1.261,38.825,0,41.403,0,44.286c0,5.367,4.351,9.718,9.719,9.718c5.368,0,9.719-
4.351,9.719-9.718");
  client.println("c0-2.943-1.312-5.574-3.378-7.355V18.436h-3.914v-2h3.914v-2.808h-4.084v-
2h4.084V8.82H11.976z M15.302,44.833");
  client.println("c0,3.083-2.5,5.583-5.583,5.583s-5.583-2.5-5.583-5.583c0-2.279,1.368-
4.236,3.326-5.104V24.257C7.462,23.01,8.472,22,9.719,22");

client.println("s2.257,1.01,2.257,2.257V39.73C13.934,40.597,15.302,42.554,15.302,44.833z'fill=#F
29C21 /></g></svg>");
client.println("</div>");
client.println("<div class='side-by-side text'>Temperature</div>");
client.println("<div class='side-by-side reading'>");
client.println(((int)temperature));
client.println("<span class='superscript'>&deg;C</span></div>");
client.println("</div>");
client.println("<div class='data pressure'>");
client.println("<div class='side-by-side icon'>");
  client.println("<svg enable-background='new 0 0 40.542 40.541'height=40.541px id=Layer_1
version=1.1 viewBox='0 0 40.542 40.541'width=40.542px x=0px xml:space=preserve
xmlns=http://www.w3.org/2000/svg xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path
d='M34.313,20.271c0-0.552,0.447-1,1-1h5.178c-0.236-4.841-2.163-9.228-5.214-12.593l-
3.425,3.424");
  client.println("c-0.195,0.195-0.451,0.293-0.707,0.293s-0.512-0.098-0.707-0.293c-0.391-0.391-
0.391-1.023,0-1.414l3.425-3.424");

```

```

client.println("c-3.375-3.059-7.776-4.987-12.634-
5.215c0.015,0.067,0.041,0.13,0.041,0.202v4.687c0,0.552-0.447,1-1,1s-1-0.448-1-1v0.25");

client.println("c0-0.071,0.026-0.134,0.041-
0.202C14.39,0.279,9.936,2.256,6.544,5.385l3.576,3.577c0.391,0.391,0.391,1.024,0,1.414");

client.println("c-0.195,0.195-0.451,0.293-0.707,0.293s-0.512-0.098-0.707-0.293L5.142,6.812c-
2.98,3.348-4.858,7.682-5.092,12.459h4.804");

client.println("c0.552,0,1,0.448,1,1s-0.448,1-
1,1H0.05c0.525,10.728,9.362,19.271,20.22,19.271c10.857,0,19.696-8.543,20.22-19.271h-5.178");

client.println("C34.76,21.271,34.313,20.823,34.313,20.271z M23.084,22.037c-0.559,1.561-
2.274,2.372-3.833,1.814");

client.println("c-1.561-0.557-2.373-2.272-1.815-3.833c0.372-1.041,1.263-1.737,2.277-
1.928L25.2,7.202L22.497,19.05");

client.println("C23.196,19.843,23.464,20.973,23.084,22.037z'fill=#26B999 /></g></svg>");
client.println("</div>");
client.println("<div class='side-by-side text'>Pressure</div>");
client.println("<div class='side-by-side reading'>");
client.println((int)pressure);
client.println("<span class='superscript'>hPa</span></div>");
client.println("</div>");
client.println("<div class='data altitude'>");
client.println("<div class='side-by-side icon'>");

client.println("<svg enable-background='new 0 0 58.422 40.639'height=40.639px id=Layer_1
version=1.1 viewBox='0 0 58.422 40.639'width=58.422px x=0px xml:space=preserve
xmlns=http://www.w3.org/2000/svg xmlns:xlink=http://www.w3.org/1999/xlink y=0px><g><path
d='M58.203,37.754l0.007-0.004l42.09,9.935l-0.001,0.001c-0.356-0.543-0.969-0.902-1.667-
0.902");

client.println("c-0.655,0-1.231,0.32-1.595,0.808l-0.011-0.007l-0.039,0.067c-0.021,0.03-
0.035,0.063-0.054,0.094L22.78,37.692l0.008,0.004");

client.println("c-0.149,0.28-0.242,0.594-
0.242,0.934c0,1.102,0.894,1.995,1.994,1.995v0.015h31.888c1.101,0,1.994-0.893,1.994-1.994");

client.println("C58.422,38.323,58.339,38.024,58.203,37.754z'fill=#955BA5 /><path
d='M19.704,38.674l-0.013-0.004l13.544-23.522L25.13,1.156l-
0.002,0.001C24.671,0.459,23.885,0,22.985,0");

client.println("c-0.84,0-1.582,0.41-2.051,1.038l-0.016-0.01L20.87,1.114c-0.025,0.039-
0.046,0.082-0.068,0.124L0.299,36.851l0.013,0.004");

```



```

client.println("C0.117,37.215,0,37.62,0,38.059c0,1.412,1.147,2.565,2.565,2.565v0.015h16.989c-
0.091-0.256-0.149-0.526-0.149-0.813");

client.println("C19.405,39.407,19.518,39.019,19.704,38.674z'fill=#955BA5 /></g></svg>");
client.println("</div>");
client.println("<div class='side-by-side text'>Altitude</div>");
client.println("<div class='side-by-side reading'>");
client.println((int)altitude);
client.println("<span class='superscript'>m</span></div>");
client.println("</div>");
client.println("</div>");
client.println("</body>");
client.println("</html>");
delay(5000);
}

```

PERHATIKAN !

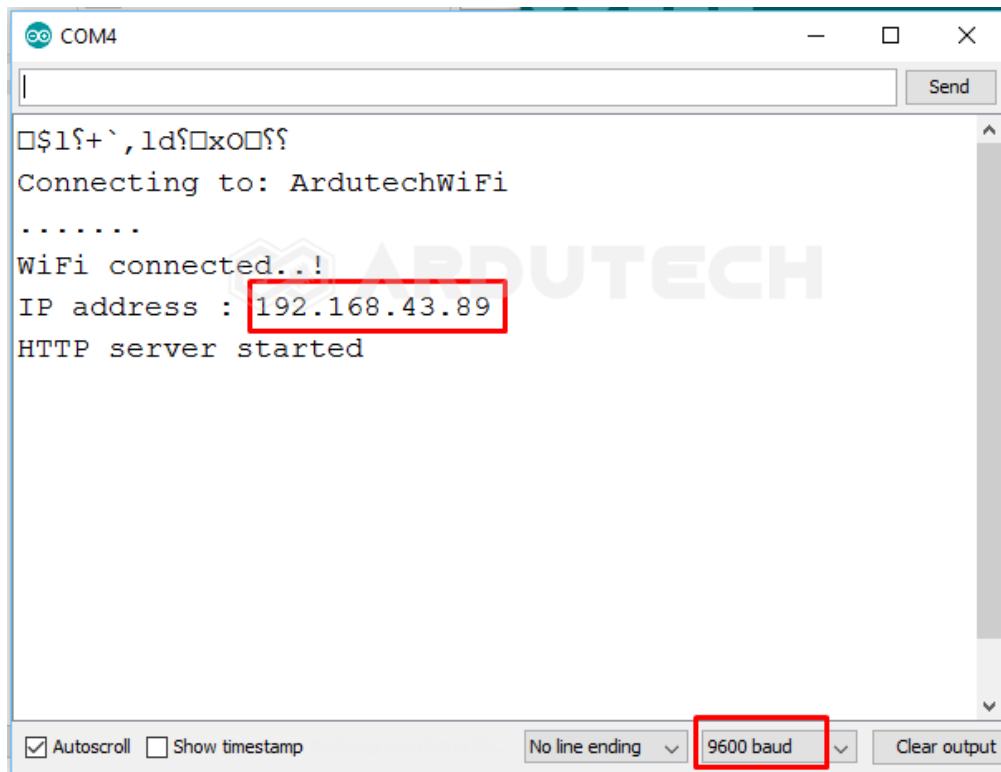
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **ssid**
- Password jaringan WiFi/hotspot : **password**

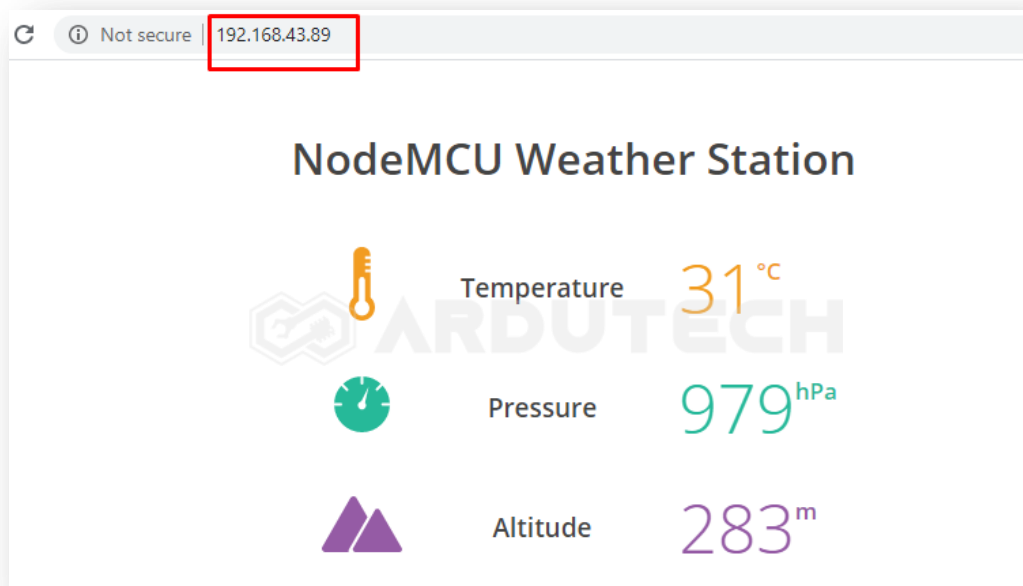
Simpan (**Save**) programnya kemudian Upload. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

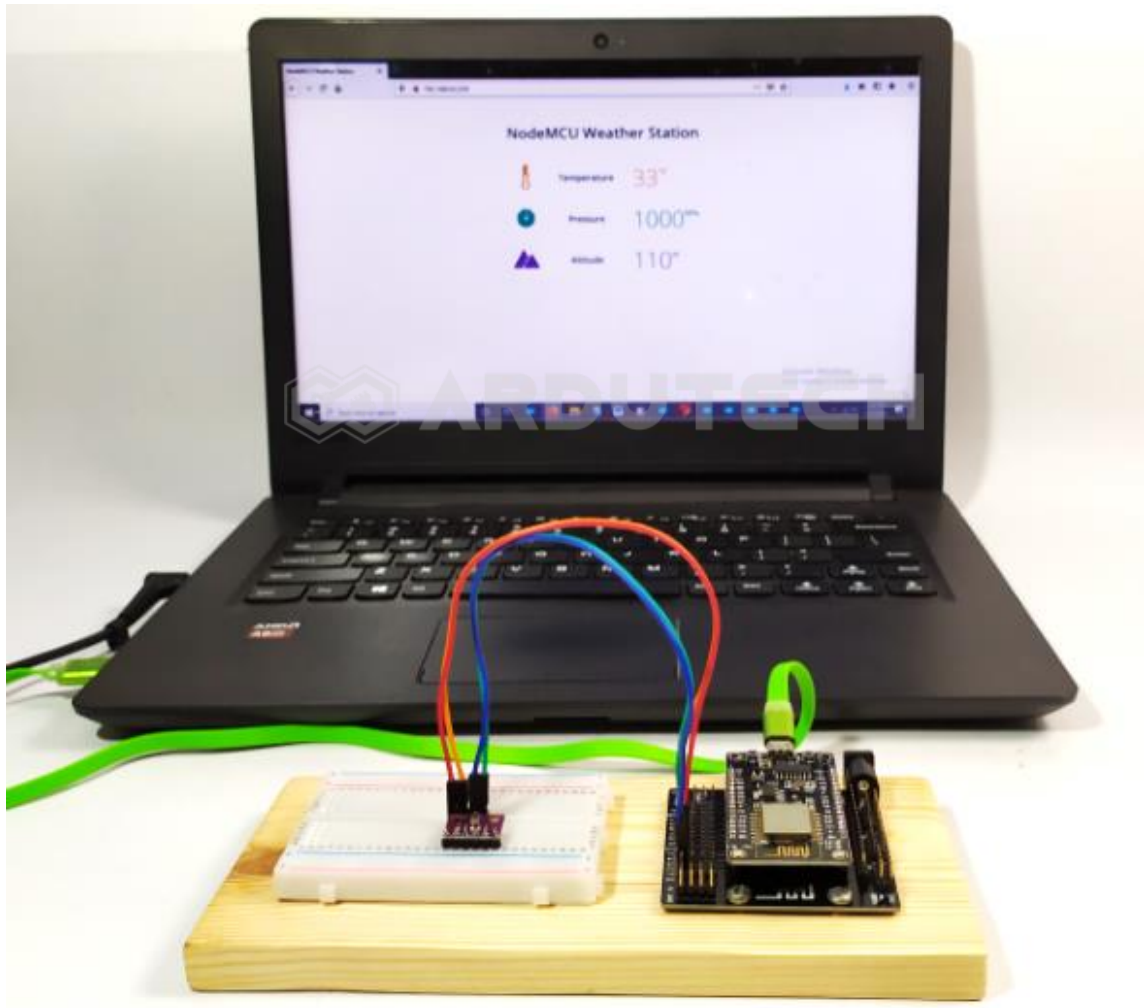
Jalannya Alat

Setelah program berhasil di Upload, silakan buka Serial Monitor dari menu **Tools → Serial Monitor**, seting baudrate pada 9600 :



Terlihat IP address pada Serial Monitor, sekarang buka di browser anda dan tuliskan IP address tadi :





Cobalah untuk memberi perubahan suhu pada sensor BMP280 dan amati perubahannya.

Selamat berkarya , semoga lancar dan bermanfaat.

Ardutech – *“Sahabat Inovasi Anda”*